

LANDSCAPES OF THE CHURCH

Overview, Shared Science & Data Architecture

Map architecture, source data, H3 hexagons, weighted summaries, shared formulas, QC principles, publication guidance, and citations

LSAP Integral Ecology Atlas | Technical Documentation | 3 October 2026

Document purpose and use

This document is the technical master for the “Landscapes of the Church” map family in the LSAP Integral Ecology Atlas. It documents the source data, scientific logic, GIS processing, field calculations, storytelling logic, quality-control checks, publication schemas, attribution, limitations, and reproducible ArcPy patterns used to create the diocesan, Episcopal Conference, and LSAP member landscape layers.

The document is intentionally long. It is meant to serve simultaneously as: (1) an internal reproducibility record; (2) a source for ArcGIS Online layer descriptions and metadata; (3) a methods appendix for academic or institutional use; (4) a handoff document for future GIS staff; and (5) a record of final analytical decisions after iterative QC.

Code appendices preserve the final analytical logic rather than every superseded debugging cell. Where a workflow evolved during QC, the documentation records the final corrected method and also preserves representative QC outputs that explain why the correction was made.

1. Map overview and analytical architecture

1.1 Map concept

“Landscapes of the Church: Land Cover and Living Systems” translates global environmental information into the spatial frameworks through which the Catholic Church and the Laudato Si’ Action Platform operate. The map does not treat ecclesiastical boundaries as ecological boundaries; rather, it uses dioceses, Episcopal Conferences, and member-centered neighborhoods as practical analytical containers for asking how human, ecological, agricultural, urban, and sparsely vegetated systems are distributed around Catholic communities and institutions.

The analytical design deliberately separates descriptive landscape structure from ecological value. A concentrated landscape is not necessarily degraded; a highly diverse land-cover mosaic is not automatically high-quality habitat; bare land can represent intact desert or exposed rock; snow and ice can represent cryospheric systems; and mapped water within an ecclesiastical polygon is contextual geography rather than a claim about canonical jurisdiction over aquatic space.

1.2 Analytical scales

Scale	Purpose
Diocese	Local-to-subnational ecclesiastical geography; 2,874 jurisdictions in the analytical source.
Episcopal Conference	National or multinational ecclesiastical coordination geography; 122 analytical conference records.
LSAP member — 20 km	Local ecological neighborhood around 2,739 geocoded, publishable LSAP member locations.
LSAP member — 100 km	Wider regional landscape context around the same 2,739 member locations; paired 1:1 using BUFFER_ID.
Environmental support geography	Esri Global Biodiversity and Conservation H3 hexagons, especially Level 5, used for global environmental summaries.

1.3 Final map hierarchy

- LSAP Members - 20 km Local Ecological Neighborhood
- LSAP Members - 100 km Regional Landscape Context
- 06 - Episcopal Conference Landscape Mosaic & Diversity
- 06 - Diocesan Landscape Mosaic & Diversity

- 05 - Episcopal Conference Sparse & Extreme Landscapes
- 05 - Diocesan Sparse & Extreme Landscapes
- 04 - Episcopal Conference Built Landscapes
- 04 - Diocesan Built Landscapes
- 03 - Episcopal Conference Green & Living Landscapes
- 03 - Diocesan Green & Living Landscapes
- 02 - Episcopal Conference Food & Working Landscapes
- 02 - Diocesan Food & Working Landscapes
- 01 - Diocesan Predominant Land Cover
- 01 - Episcopal Conference Predominant Land Cover
- 00 - Diocesan Land-Cover Composition
- 00 - Episcopal Conference Land-Cover Composition

1.4 Analytical series

ID	Analysis	Core question
00	Land-Cover Composition	Full eight-class composition and areas; foundational descriptive layer.
01	Predominant Land Cover	Leading and second land-cover systems; dominance gap and composition class.
02	Food & Working Landscapes	Cropland + rangeland combined; balance and primary working-landscape type.
03	Green & Living Landscapes	Trees + other vegetation; balance and leading green-cover type.
04	Built Landscapes	Built-up land combined with WorldPop 2025/2030 population change.
05	Sparse & Extreme Landscapes	Bare land + snow/ice; descriptive extent and balance.
06	Landscape Mosaic & Diversity	Normalized Shannon diversity, effective land-cover types, leading pair, coverage, mosaic context.
Member	20 km / 100 km	Member-centered local-to-regional comparison with transition narratives and coverage-aware interpretation.

2. Source data, provenance, licensing, and attribution

2.1 Catholic ecclesiastical geography

The ecclesiastical geography used in this map is a project-maintained Catholic jurisdictional framework. Diocesan geometry and enrichment were developed through the GoodLands / Ecclesiastical Geography program and subsequently maintained as the project's authoritative v2.0 jurisdictional layer. Enrichment draws on Holy See statistical publications and Catholic-Hierarchy.org, with versioned provenance retained in the jurisdictional database.

The analytical baseline for this map uses 2,874 diocesan-scale records and 122 Episcopal Conference records. Conference geographies are project-derived analytical units used for aggregate comparison; they should not be interpreted as replacing official canonical descriptions of competence or territorial scope.

Primary project layer	Dioceses of the Catholic Church v2.0 / internal enriched derivative used for analysis
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Diocesan analytical source	LSAP_Diocese_BiodiversitySummary
Conference analytical source	LSAP_EpiscopalConferences_BiodiversitySummary
Statistical enrichment	Annuario Pontificio (multiple years), Annuarium Statisticum Ecclesiae / Holy See statistical sources, Catholic-Hierarchy.org
Project provenance	Versioned jurisdictional keys, boundary provenance, source dates, enrichment provenance, and terms/attribution maintained in the project metadata system.

2.2 Laudato Si' Action Platform member data

The member layers use the publishable/geocodable subset of Laudato Si' Action Platform registrations. The analytical member products documented here contain 2,739 point records at both 20 km and 100 km scales. The two scales are paired exactly 1:1 using BUFFER_ID. Account_Name is not a safe join key because duplicate names occur.

Public-facing descriptions should continue to state that member points represent a subset of LSAP members who permitted mapping and had sufficient geocodable information. The layer should not be described as the complete universe of all LSAP participants.

2.3 Environmental source: Esri Global Biodiversity and Conservation Hexagons

The environmental foundation is Esri's Global Hexagons for Biodiversity and Conservation in ArcGIS Living Atlas of the World. Esri describes the product as terrestrial and marine biodiversity/conservation information summarized into nested H3 hexagons at Levels 2–5. The 2026 update expanded the product to more than 80 summary statistics from 28 environmental datasets; the earlier 2025 release summarized more than 20 datasets. Level 5 is approximately 250–253 km² per hexagon.

The product includes, among other sources, the World Database of Protected Areas, SBTN conservation hotspots, biodiversity hotspots, biodiversity intactness, IUCN Red List indicators, iNaturalist observations, Esri Sentinel-2 Land Use/Land Cover, World Terrestrial Ecosystems, land-cover vulnerability to 2050, tree canopy, carbon/biomass indicators, terrain, Global Forest Watch, and WorldPop population.

Authoritative source links: [Esri 2026 update](#) | [Living Atlas group-layer overview](#) | [Hexagonal Earth StoryMap](#)

Licensing note: the ArcGIS item describes the overall work as CC BY 4.0 and identifies IUCN Red List-derived variables as CC BY-NC 4.0. Because Living Atlas items can be updated, final ArcGIS Online publication should retain the credit string and license language attached to the exact item/version used.

2.4 Key Level-5 environmental indicators used across the atlas

- WorldPop population estimates for 2025 and 2030.
- Protected-area coverage and distance/context from WDPA-derived summaries.
- IUCN Red List richness and threatened-species richness indicators.
- iNaturalist observations and citizen-science context.
- Esri Sentinel-2 land-use/land-cover shares: Water, Trees, Cropland, Built, Rangeland, Bare, Snow/Ice, and Other Vegetation; Cloud and NoData retained only for coverage accounting.
- World Terrestrial Ecosystems and projected ecosystem / land-cover vulnerability toward 2050.
- Biodiversity intactness and conservation-hotspot indicators.
- Tree canopy and carbon / biomass indicators.
- Terrain indicators including elevation and slope.
- Global Forest Watch change indicators and related environmental context.

3. Esri biodiversity and conservation H3 hexagons

3.1 Why H3 hexagons

H3 provides a hierarchical, spatially indexed tessellation. Esri’s environmental hexagon product uses multiple H3 resolutions so the same environmental themes can be explored consistently from global to regional scales. Level 5 was selected for the LSAP analytical workflow because it is fine enough to preserve meaningful environmental variation while remaining computationally tractable for thousands of dioceses, conference territories, and member-centered buffers.

Esri reports that the hexagon layers were created in ArcGIS Pro using the H3 option in Generate Tessellation. The Living Atlas product stores precomputed summary statistics such as means, sums, majority classes, and percent-area variables. This means the LSAP analysis is not repeatedly resampling the original global rasters; instead, it performs a second-stage spatial summarization from a standardized analysis-ready H3 geography.

3.2 Level-5 size and interpretation

A Level-5 hexagon is approximately 250–253 km², although exact polygon area varies slightly with H3 geometry and latitude. Values stored on each hexagon summarize the source environmental datasets for that cell. When a Church geography cuts through a hexagon, the project uses the area of the overlapping hexagon fragment to weight rate-like or mean-like indicators rather than treating a 1% sliver as equivalent to a 100% overlap.

3.3 Summarize Within / overlay logic

ArcGIS Pro’s Summarize Within tool overlays a summary polygon layer with features to calculate geometry and attribute statistics inside each polygon. Esri’s documentation also describes geographically weighted statistics for area features, where weights reflect the proportion of summarized geometry falling within the summary polygon. For the LSAP workflow, this principle is made explicit through an intersect-and-weight procedure whenever transparent per-fragment weighting is required.

ArcGIS Pro documentation: Summarize Within (Analysis)

3.4 Project workflow: hexagons → Church / member geography

1. Begin with the Esri Global Biodiversity and Conservation Level-5 H3 polygon layer and the target analysis geography (dioceses, Episcopal Conferences, 20 km member buffers, or 100 km member buffers).
2. Run a polygon overlay (Summarize Within or Intersect, depending on the statistic and need for explicit fragment-level control).
3. For explicit weighting, create a fragment area field using geodesic area in square kilometers.
4. Retain the target identifier (for example Diocese key, Conference key, or BUFFER_ID), the source H3 identifier, the source environmental variables, and overlap area.
5. For continuous means, rates, percentages, densities, or index values, calculate an overlap-area-weighted mean using only non-null fragments.
6. For categorical variables, determine the category associated with the greatest valid overlapping area or use the source majority/mode where analytically appropriate.
7. For area totals, sum the relevant overlap-area-derived quantity.
8. For count/amount variables that represent totals for an entire hexagon, prorate by overlap fraction only when the assumption is scientifically defensible; otherwise retain them as contextual source totals and label the limitation.
9. Join the summarized results back to the target geography using the stable target key.
10. Run coverage, null, unit, and plausibility QC before deriving publication layers.

3.5 Area-weighted mean

Area-weighted mean	$\bar{x}_w = \sum(x_i \times A_i) / \sum(A_i)$ x_i is the source hexagon value and A_i is the area of the
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	hexagon fragment overlapping the target geography. The denominator includes only fragments with valid x_i values.
Overlap fraction	$f_i = A_{\text{overlap},i} / A_{\text{hex},i}$ Used when a hexagon-level amount must be proportioned to the fraction of the source hexagon represented inside the target geography.
Prorated amount	$T_{\text{target}} \approx \sum(T_i \times f_i)$ Use only for count/amount variables for which spatial prorating is a reasonable approximation. Do not apply this formula blindly to rates, indices, or categorical variables.

3.6 Aggregation policy by field type

Field type	Preferred aggregation	Examples / notes
Continuous mean / index	Overlap-area-weighted mean	Biodiversity intactness, carbon density, mean canopy, mean slope/elevation, vulnerability indices.
Percentage / proportion	Overlap-area-weighted mean	Protected-area %, land-cover %, hotspot %, other percent fields.
Area	Sum of intersected or derived area	Area of protected land, area by land-cover class.
Count / amount	Prorated sum only when defensible; otherwise contextual	Population or other hex-wide totals require explicit assumptions.
Categorical class	Largest valid overlap / dominant area	Major ecosystem / land-cover class.
Distance	Area-weighted mean or target-specific summary	Distance-to-protected-area-style contextual variables; method documented field-by-field.

4. Shared land-cover science and analytical conventions

4.1 Eight valid land-cover classes

Class	Field	Interpretation
Water	PCT_LULC_Water	Mapped water within the analysis geography; treated as environmental context rather than canonical aquatic jurisdiction.
Trees	PCT_LULC_Trees	Tree-dominated land cover.
Cropland	PCT_LULC_Crops	Cultivated / agricultural cropland.
Built	PCT_LULC_Built	Built-up / developed land.
Rangeland	PCT_LULC_Rangeland	Rangeland / grassland-oriented working landscapes in the source classification.
Bare	PCT_LULC_Bare	Bare or sparsely vegetated surfaces; not inherently degraded.
Snow/Ice	PCT_LULC_SnowIce	Snow- and ice-dominated cover.
Other Vegetation	PCT_LULC_Veg	Other mapped vegetation not represented in the tree/crop/rangeland classes.

4.2 Cloud and NoData are coverage, not landscape classes

PCT_LULC_Cloud and PCT_LULC_NoData are excluded from ecological composition and diversity calculations. They are retained only for data-coverage accounting. This decision prevents missing or obscured source pixels from being interpreted as a real landscape system.

Valid classified coverage	$VALID_COVER_PCT = \sum \text{eight valid PCT_LULC classes}$ The eight classes above are summed as the portion of the analysis geography represented by meaningful land-cover classes.
Unclassified coverage	$UNCLASSIFIED_PCT = \max(0, 100 - VALID_COVER_PCT)$ Used in coverage-aware narratives and QC.
Member source residual	$OTHER_UNMAPPED_PCT = \max(0, 100 - (VALID_COVER_PCT + Cloud + NoData))$ Needed because some 100 km member records contained additional unrepresented source coverage beyond Cloud + NoData.

4.3 Unit normalization

A major QC lesson was that analytical area fields inherited from the hexagon summaries were stored at square-meter scale even when later publication labels were intended to be square kilometers. Every derivative layer therefore diagnoses units before conversion. Area fields are divided by 1,000,000 exactly once, and the code refuses to reconvert values that already appear to be in km².

Area conversion	$Area_km^2 = Area_m^2 / 1,000,000$ Applied only after a median-magnitude diagnostic. This protects against double conversion.
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4.4 Predominance

Leading and second land-cover types are identified by sorting the eight valid class shares from largest to smallest. TOP_PCT and SECOND_PCT are retained as percentages of the complete analysis geography. The leading–second gap is the difference in percentage points and is used as a measure of dominance.

Leading gap	$TOP_GAP = TOP_PCT - SECOND_PCT$ A larger gap indicates stronger dominance by the leading land-cover type.
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4.5 Normalized Shannon diversity and effective land-cover types

Analysis 06 and the member landscapes normalize the eight valid land-cover shares to the classified portion of the geography before calculating compositional diversity. This is essential when Cloud/NoData or other unclassified coverage is nontrivial.

Normalized class proportion	$p_i = c_i / \sum(c_i)$ c_i is the raw percentage of the complete geography for valid land-cover class i .
Shannon diversity	$H = -\sum(p_i \times \ln(p_i))$ Calculated across the eight valid land-cover classes with $p_i > 0$.

Normalized diversity	MOSAIC_DIV = $100 \times H / \ln(8)$ Maps Shannon diversity onto a 0–100 scale.
Effective number of types	EFFECTIVE_TYPES = $\exp(H)$ Interpretable as the number of equally common land-cover types that would produce the observed Shannon diversity.

MEANINGFUL_TYPES counts normalized classified classes $\geq 1\%$; MAJOR_TYPES counts normalized classified classes $\geq 5\%$. The diversity metrics describe composition and evenness only. They are not measures of species diversity, habitat quality, ecological health, or conservation value.

4.6 Mosaic typology

Class	Rule
Highly Diverse Mosaic	MOSAIC_DIV ≥ 65 ; EFFECTIVE_TYPES ≥ 4.0 ; MEANINGFUL_TYPES ≥ 5 ; MAJOR_TYPES ≥ 4 ; leading gap within classified cover ≤ 15 pp.
Diverse Mosaic	MOSAIC_DIV ≥ 55 ; EFFECTIVE_TYPES ≥ 3.25 ; MEANINGFUL_TYPES ≥ 4 ; MAJOR_TYPES ≥ 3 ; leading gap within classified cover ≤ 35 pp.
Concentrated Landscape	MOSAIC_DIV < 35 OR EFFECTIVE_TYPES < 2.0 OR MAJOR_TYPES ≤ 1 OR leading gap within classified cover ≥ 60 pp.
Structured Mosaic	All remaining cases.

These classes are descriptive analytical labels, not environmental grades or rankings.

4.7 Rank and percentile convention

Where rank fields were retained in Analyses 02–05, the analytical rank uses competition ranking with rank 1 assigned to the highest value. Percentile represents the percentage of observations at or below the observed value. Ranks are scale-specific: dioceses are compared only with dioceses; Episcopal Conferences only with Conferences. Rank/percentile language was intentionally removed from later story prose where it made popup text feel overly statistical.

14. Cross-scale storytelling and popup design

14.1 Storytelling principles

- Put the place first. Popup prose should explain where the member or Church geography sits, not simply restate a table.
- Describe composition before interpretation.
- Avoid rank-heavy prose; keep rank fields analytical where useful but prefer direct landscape language.
- Do not describe bare land, snow/ice, concentrated composition, or high built share as inherently bad.
- Do not equate land-cover diversity with biodiversity, habitat quality, ecological health, or conservation value.
- Treat Water as surrounding physical geography and watershed/coastal context, not an assertion of ecclesiastical jurisdiction over water.
- When classified coverage is incomplete, explicitly say that the composition refers to the classified portion.
- At member scale, emphasize the transition from local 20 km neighborhood to wider 100 km region.

14.2 Member popup narrative model

The most distinctive storytelling innovation in the member layers is the local-to-regional transition. Instead of showing two independent percentage tables, the popup explains how the landscape changes as the spatial context expands.

Example pattern: “Built-up land leads around the member locally, while cropland becomes the leading land-cover type across the wider regional context. Within the classified landscape, cropland increases while built-up land decreases as the view expands.”

This makes the map actionable without prescribing a particular intervention. A locally urban institution can see its immediate heat/tree-canopy/stormwater context while also understanding its relationship to a broader agricultural, forest, rangeland, or watershed system.

14.3 Stewardship-language library

Leading type	Story themes
Built	urban heat; tree canopy; stormwater; habitat in developed areas; mobility; access to everyday nature
Trees	forest/woodland stewardship; habitat connectivity; canopy protection; watershed function; long-term carbon storage
Cropland	soil health; agroecology; pollinators; water quality; food systems; agricultural-community relationships
Rangeland	grassland/grazing systems; soil cover; drought resilience; connectivity; fire-aware stewardship
Water	watersheds; water quality; flood/coastal resilience; shoreline habitats; land-water relationships
Bare	erosion; water scarcity; fragile soils; arid-land ecology; stewardship suited to sparse vegetation
Snow/Ice	seasonal water systems; mountain/polar ecology; cryosphere dynamics; climate sensitivity
Other Vegetation	shrub/herbaceous habitats; fire ecology; soil condition; watershed function; connectivity

15. Quality assurance, limitations, and interpretation rules

15.1 Core QC philosophy

The final workflow intentionally uses one substantive scientific/distribution QC per analysis plus light completeness/content checks. Early iterations used repeated large QC scripts; the final process reduces redundancy while preserving the checks most likely to catch scientific or publication errors.

- Feature-count checks before and after derivation.
- Required-field checks before calculations.
- Unit diagnostics before any area conversion.
- Formula/range checks for impossible percentages, negative values, and diversity outside 0–100.
- Coverage checks separating valid classes from Cloud/NoData and other missing source coverage.
- Distribution checks for new derived metrics and typologies.
- Story completeness checks (missing story fields = 0).
- Spot checks of representative high/low/extreme cases.
- Publication-schema checks after field deletion.
- Map-placement checks to confirm the intended layer order.

15.2 Important limitations

- Ecclesiastical analytical polygons are not ecological boundaries; ecological systems cross jurisdictional borders.
- Episcopal Conference polygons are analytical coordination geographies and may include complex multi-country, overlapping, or special territorial situations.
- Land-cover summaries inherit the source resolution, classification, vintage, and limitations of the Living Atlas inputs.
- H3-weighted summaries are approximations when the original source variable is itself a hexagon summary rather than a continuous surface.
- Prorating counts or amounts across partial hexagons assumes spatial distribution within the source hexagon; this should be avoided or flagged when the assumption is weak.
- WorldPop and Catholic population values are independent estimates with different sources and methodologies.
- Land-cover diversity is compositional diversity, not biological diversity.
- Member points represent the consented/geocodable LSAP subset and are not a census of all LSAP participants.
- The 100 km member analysis geography is the source workflow's represented analysis area and should not be described as a perfect theoretical 100 km circle in every case.
- Water is mapped physical context and should not be used to imply canonical jurisdiction over aquatic areas.

15.3 Versioning and Living Atlas updates

Esri released an improved Global Biodiversity and Conservation Hexagons product in July 2026 and indicated that the older 2025 version would retire in December 2026. The project should record the exact ArcGIS item ID, access date, schema version, and source credits used for each analytical refresh. If a future update changes field definitions or source vintages, rerun the hexagon and land-cover QC before comparing results longitudinally.

16. ArcGIS Online publication guidance

16.1 Minimum metadata block for every layer

- Title / layer alias matching the final map name.
- One-paragraph summary written for a nontechnical audience.
- Long description with analytical scale, method, valid classes, and coverage caveat.
- Credits / attribution for ecclesiastical or LSAP source data plus Esri/Living Atlas environmental sources.
- Citation text and source access/version date.
- Terms of use and any source-specific license restrictions, especially IUCN-derived fields.
- Field aliases in plain language.
- Methodology note for area-weighted summaries and normalized classified-cover diversity.
- Limitations / disclaimer distinguishing land-cover diversity from biodiversity and jurisdiction from ecological extent.
- Tags including LSAP, Laudato Si', Catholic Church, Integral Ecology, Land Cover, H3, Biodiversity, Conservation, and the relevant geographic scale.

16.2 Recommended public description template

This layer is part of the LSAP Integral Ecology Atlas “Landscapes of the Church” map series. It summarizes land-cover and environmental context for Catholic ecclesiastical geographies or publishable Laudato Si' Action Platform member locations using analysis-ready environmental data from ArcGIS Living Atlas. Land-cover composition is based on eight mapped classes: Water, Trees, Cropland, Built, Rangeland, Bare, Snow/Ice, and Other Vegetation.

Cloud and NoData are treated as coverage limitations rather than landscape classes. Where environmental hexagons intersect only part of an analysis geography, continuous/rate-like indicators are summarized using overlap-area weighting. Derived landscape-diversity metrics describe land-cover composition and evenness, not species diversity or ecological quality.

Reference code: environmental hexagon summarization

A.1 Hexagon intersection and overlap-area-weighted summaries

```
import arcpy
import os
from collections import defaultdict

GDB = r"C:\path\to\project.gdb"

TARGET = os.path.join(GDB, "Target_Geography")      # dioceses, conferences, or buffers
HEX = os.path.join(GDB, "BiodiversityHexagons_L5")
INTERSECT = os.path.join(GDB, "Target_Hex_Intersect")

TARGET_ID = "TARGET_ID"
HEX_ID = "GRID_ID"

# Example rate/mean fields.
WEIGHTED_FIELDS = [
    "PCT_LULC_Trees",
    "PCT_LULC_Crops",
    "PCT_LULC_Built",
    "PCT_PROTECTED_AREA",
    "Biodiversity_Intactness",
    "Tree_Canopy",
    "Carbon_Density",
    "Slope_Mean",
    "Elevation_Mean"
]

arcpy.analysis.Intersect(
    [TARGET, HEX],
    INTERSECT,
    join_attributes="ALL"
)

# Use geodesic area when working globally.
if "OVERLAP_KM2" not in {f.name for f in arcpy.ListFields(INTERSECT)}:
    arcpy.management.AddField(INTERSECT, "OVERLAP_KM2", "DOUBLE")

arcpy.management.CalculateGeometryAttributes(
    INTERSECT,
    [["OVERLAP_KM2", "AREA_GEODESIC"]],
    area_unit="SQUARE_KILOMETERS"
)

# Numerator and denominator are maintained separately for EACH field so nulls
# do not dilute the denominator.
num = defaultdict(lambda: defaultdict(float))
den = defaultdict(lambda: defaultdict(float))

fields = [TARGET_ID, "OVERLAP_KM2"] + WEIGHTED_FIELDS

with arcpy.da.SearchCursor(INTERSECT, fields) as cur:
    for row in cur:
        target_id = row[0]
        overlap = float(row[1] or 0)

        if overlap <= 0:
            continue

        for i, fld in enumerate(WEIGHTED_FIELDS, start=2):
            value = row[i]
```

```

        if value is None:
            continue

        num[target_id][fld] += float(value) * overlap
        den[target_id][fld] += overlap

weighted = defaultdict(dict)

for target_id in num:
    for fld in WEIGHTED_FIELDS:
        if den[target_id][fld] > 0:
            weighted[target_id][fld] = (
                num[target_id][fld] /
                den[target_id][fld]
            )
        else:
            weighted[target_id][fld] = None

# Join/write these weighted values back to a copy of TARGET using TARGET_ID.

```

A.2 Optional prorating of a hexagon-wide amount

Use only when the source variable is a COUNT/AMOUNT and prorating is
scientifically defensible. Do not use this blindly for rates or indices.

```

# INTERSECT must contain:
#   OVERLAP_KM2
#   HEX_AREA_KM2
#   SOURCE_TOTAL

prorated = defaultdict(float)

with arcpy.da.SearchCursor(
    INTERSECT,
    [TARGET_ID, "OVERLAP_KM2", "HEX_AREA_KM2", "SOURCE_TOTAL"]
) as cur:

    for target_id, overlap, hex_area, source_total in cur:

        if (
            overlap is None or
            hex_area is None or
            source_total is None or
            hex_area <= 0
        ):
            continue

        fraction = min(1.0, max(0.0, overlap / hex_area))

        prorated[target_id] += (
            float(source_total) *
            fraction
        )

```

A.3 Shared land-cover helpers

```

import math

LANDCOVER = {
    "Water": "PCT_LULC_Water",
    "Trees": "PCT_LULC_Trees",
    "Cropland": "PCT_LULC_Crops",
    "Built": "PCT_LULC_Built",
    "Rangeland": "PCT_LULC_Rangeland",
    "Bare": "PCT_LULC_Bare",
    "Snow/Ice": "PCT_LULC_SnowIce",
    "Other Vegetation": "PCT_LULC_Veg"
}

def safe(v):
    return 0.0 if v is None else float(v)

def normalized_classified(raw):

```

```

    valid = sum(max(0.0, safe(v)) for v in row.values())

    if valid <= 0:
        return valid, {k: 0.0 for k in row}

    return (
        valid,
        {k: 100.0 * max(0.0, safe(v)) / valid for k, v in row.items()}
    )

def shannon_metrics(row):
    valid, pct = normalized_classified(row)

    props = {k: v / 100.0 for k, v in pct.items()}

    H = -sum(
        p * math.log(p)
        for p in props.values()
        if p > 0
    )

    diversity = 100.0 * H / math.log(8)
    effective = math.exp(H)

    meaningful = sum(v >= 1.0 for v in pct.values())
    major = sum(v >= 5.0 for v in pct.values())

    ordered = sorted(
        row.items(),
        key=lambda kv: safe(kv[1]),
        reverse=True
    )

    top_type, top_pct = ordered[0]
    second_type, second_pct = ordered[1]

    classified_gap = (
        pct[top_type] -
        pct[second_type]
    )

    return {
        "valid_cover": valid,
        "diversity": diversity,
        "effective": effective,
        "meaningful": meaningful,
        "major": major,
        "top_type": top_type,
        "top_pct": safe(top_pct),
        "second_type": second_type,
        "second_pct": safe(second_pct),
        "classified_gap": classified_gap
    }

def mosaic_context(m):
    if (
        m["diversity"] >= 65 and
        m["effective"] >= 4.0 and
        m["meaningful"] >= 5 and
        m["major"] >= 4 and
        m["classified_gap"] <= 15
    ):
        return "Highly Diverse Mosaic"

    if (
        m["diversity"] >= 55 and
        m["effective"] >= 3.25 and
        m["meaningful"] >= 4 and
        m["major"] >= 3 and
        m["classified_gap"] <= 35
    ):
        return "Diverse Mosaic"

    if (

```

```

    m["diversity"] < 35 or
    m["effective"] < 2.0 or
    m["major"] <= 1 or
    m["classified_gap"] >= 60
  ):
    return "Concentrated Landscape"

return "Structured Mosaic"

```

Master QC snapshot

Appendix B. Consolidated retained QC logs

This appendix brings together the most important executed QC outputs preserved from the analytical workflow. It is not every diagnostic printed during development; it is the subset that materially supports the final scientific and publication decisions.

Retained QC output

```

MASTER FEATURE COUNTS
Diocesan analytical geography:      2,874
Episcopal Conference analytical units: 122
LSAP member 20 km derivative:      2,739
LSAP member 100 km derivative:     2,739
Member pairing key: BUFFER_ID
20 km <-> 100 km unmatched keys:    0

```

Retained QC output

```

MEMBER STORY COMPLETENESS
20 KM
STORY_NEIGHBORHOOD      missing=0
STORY_LOCAL_LANDSCAPE   missing=0
STORY_LOCAL_TO_REGION   missing=0
STORY_STEWARDSHIP       missing=0
STORY_DATA_NOTE         missing=0
STORY_SUMMARY           missing=0
STORY_COMPREHENSIVE     missing=0

```

```

100 KM
STORY_REGION            missing=0
STORY_REGIONAL_LANDSCAPE missing=0
STORY_LOCAL_TO_REGION   missing=0
STORY_STEWARDSHIP       missing=0
STORY_DATA_NOTE         missing=0
STORY_SUMMARY           missing=0
STORY_COMPREHENSIVE     missing=0

```

Retained QC output

```

MEMBER FINAL MAP PLACEMENT
LSAP Members - 20 km Local Ecological Neighborhood
LSAP Members - 100 km Regional Landscape Context
06 - Episcopal Conference Landscape Mosaic & Diversity
06 - Diocesan Landscape Mosaic & Diversity
05 - Episcopal Conference Sparse & Extreme Landscapes
05 - Diocesan Sparse & Extreme Landscapes
04 - Episcopal Conference Built Landscapes
04 - Diocesan Built Landscapes
03 - Episcopal Conference Green & Living Landscapes
03 - Diocesan Green & Living Landscapes
02 - Episcopal Conference Food & Working Landscapes
02 - Diocesan Food & Working Landscapes
01 - Diocesan Predominant Land Cover
01 - Episcopal Conference Predominant Land Cover
00 - Diocesan Land-Cover Composition
00 - Episcopal Conference Land-Cover Composition

```

Appendix C. Citations, credits, and attribution text

C.1 Esri / Living Atlas

- Esri. (2026, July 8). Global Biodiversity and Conservation Hexagons... but better! ArcGIS Blog. <https://www.esri.com/arcgis-blog/products/arcgis-living-atlas/sustainable-development/global-biodiversity-and-conservation-hexagons-but-better>
- Pisut, D., & VanGraafeiland, K. (2025, July 9). Introducing Global Biodiversity and Conservation Hexagons. ArcGIS Blog. <https://www.esri.com/arcgis-blog/products/arcgis-living-atlas/sustainable-development/biodiversity-conservation-hexagons>
- Esri. Global Hexagons for Biodiversity and Conservation [ArcGIS Living Atlas group layer]. Project-accessed version, 2026. <https://www.arcgis.com/home/item.html?id=f1bdbdcac3c54ad8b5b21f5b17d91e8d>
- Esri. Summarize Within (Analysis)—ArcGIS Pro documentation. <https://pro.arcgis.com/en/pro-app/3.6/tool-reference/analysis/summarize-within.htm>

Living Atlas licensing: use the exact item's current credit string. The documented group layer identifies the overall work as CC BY 4.0 and IUCN Red List-derived variables as CC BY-NC 4.0. Attribution should include the source organizations listed by Esri, including Esri, UNEP/WCMC, SBTN, National Geographic, Conservation International, IUCN, iNaturalist, ESA, ETH Zurich, Global Fishing Watch, WorldPop, and Hansen/UMD/Google/USGS/NASA, as applicable to the fields used.

C.2 Ecclesiastical geography and Church statistics

Project boundary citation (historical foundation): Burhans, M., Bell, J., Burhans, D., Carmichael, R., Cheney, D., Deaton, M., Emge, T., Gerlt, B., Grayson, J., Herries, J., Keegan, H., Skinner, A., Smith, M., Sousa, C., & Trubetskoy, S. Diocesan Boundaries of the Catholic Church [Feature Layer], v1.0. The Landscapes of the Church analyses use the subsequently revised and enriched project v2.0 jurisdictional framework; cite the exact project version and provenance record distributed with the layer.

Holy See. Annuario Pontificio [Pontifical Yearbook]. Multiple annual editions used in the project. Vatican City: Libreria Editrice Vaticana / Holy See.

Holy See / Central Office of Church Statistics. Annuario Statisticum Ecclesiae [Statistical Yearbook of the Church]. Multiple annual editions digitized and aligned to historical ecclesiastical geographies in the project database.

Cheney, D. M. Catholic-Hierarchy.org. Ecclesiastical-jurisdiction and bishop/ordinary enrichment used by the project; accessed September 2026. Exact access dates should be retained in the project provenance fields.

C.3 Laudato Si' Action Platform

Laudato Si' Action Platform / Dicastery for Promoting Integral Human Development. LSAP member and engagement data, project extract used in 2026 analyses. Public member layers include only records with permission for mapping and sufficient geocodable information.

C.4 Suggested compact ArcGIS Online credit

Ecclesiastical geography and Church statistics: LSAP Integral Ecology Atlas / project Catholic jurisdictional framework; Holy See statistical publications; Catholic-Hierarchy.org. Environmental context: Esri ArcGIS Living Atlas Global Biodiversity and Conservation Hexagons and underlying credited source organizations. LSAP member layers: Laudato Si' Action Platform / Dicastery for Promoting Integral Human Development. See item description for field-specific attribution and licensing.

C.5 Documentation note

This technical master should be updated whenever a new Living Atlas hexagon version, jurisdictional boundary version, Church-statistics vintage, or LSAP member extract is used. For reproducibility, preserve the ArcGIS item ID, access date, field schema, geodatabase feature-class names, and final QC output with each release.